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Methylomonas rivi sp. nov., *Methylomonas rosea* sp. nov., *Methylomonas aurea* sp. nov. and *Methylomonas subterranea* sp. nov., type I methane-oxidizing bacteria isolated from a freshwater creek and the deep terrestrial subsurface

Christopher A. Abin[1,2,*], Christopher T. Garner[1], Krithivasan Sankaranarayanan[1,2], Reid A. Sindelar[1], Kyrah F. Kotary[1], Rosa M. Garner[1], Samantha C. Barclay[1], Haiyuan Cai[3], Paul A. Lawson[1] and Lee R. Krumholz[1]

Abstract

Four methane-oxidizing bacteria, designated as strains WSC-6^T, WSC-7^T, SURF-1^T, and SURF-2^T, were isolated from Saddle Mountain Creek in southwestern Oklahoma, USA, and the Sanford Underground Research Facility (SURF) in Lead, South Dakota, USA. The strains were Gram-negative, motile, short rods that possessed intracytoplasmic membranes characteristic of type I methanotrophs. All four strains were oxidase-negative and weakly catalase-positive. Colonies ranged from pale pink to orange in colour. Methane and methanol were the only compounds that could serve as carbon and energy sources for growth. Strains WSC-6^T and WSC-7^T grew optimally at lower temperatures (25 and 20 °C, respectively) compared to strains SURF-1^T and SURF-2^T (40 °C). Strains WSC-6^T and SURF-2^T were neutrophilic (optimal pH of 7.5 and 7.3, respectively), while strains WSC-7^T and SURF-1^T were slightly alkaliphilic, with an optimal pH of 8.8. The strains grew best in media amended with ≤0.5% NaCl. The major cellular fatty acids were $C_{14:0}$, $C_{16:1}\,\omega 8c$, $C_{16:1}\,\omega 7c$, and $C_{16:1}\,\omega 5c$. The DNA G+C content ranged from 51.5 to 56.0 mol%. Phylogenetic analyses indicated that the strains belonged to the genus *Methylomonas*, with each exhibiting 98.6–99.6% 16S rRNA gene sequence similarity to closely related strains. Genome-wide estimates of relatedness (84.5–88.4% average nucleotide identity, 85.8–92.4% average amino acid identity and 27.4–35.0% digital DNA–DNA hybridization) fell below established thresholds for species delineation. Based on these combined results, we propose to classify these strains as representing novel species of the genus *Methylomonas*, for which the names *Methylomonas rivi* (type strain WSC-6^T=ATCC TSD-251^T=DSM 112293^T), *Methylomonas rosea* (type strain WSC-7^T=ATCC TSD-252^T=DSM 112281^T), *Methylomonas aurea* (type strain SURF-1^T=ATCC TSD-253^T=DSM 112282^T), and *Methylomonas subterranea* (type strain SURF-2^T=ATCC TSD-254^T=DSM 112283^T) are proposed.

INTRODUCTION

The genus *Methylomonas* comprises type I methanotrophs that belong to the family *Methylococcaceae* of the class *Gammaproteobacteria*. The type species of this genus, *Methylomonas methanica*, was the first methanotroph to be isolated in pure culture and was originally named '*Bacillus methanicus*' [1]. In addition to *M. methanica*, there are nine members of the genus with validly published names, as follows: *M. albis* [2], *M. aurantiaca* [3], *M. defluvii* [4], *M. fluvii* [2], *M. fodinarum* [3], *M. koyamae* [5], *M. lenta* [6], *M. paludis* [7], *M. rapida* [8], and *M. scandinavica* [9]. Four other species, '*M. denitrificans*' [10], '*M. montana*' [11],

Author affiliations: [1]School of Biological Sciences, University of Oklahoma, Norman, Oklahoma, USA; [2]Laboratories of Molecular Anthropology and Microbiome Research, University of Oklahoma, Norman, Oklahoma, USA; [3]Nanjing Institute of Geography and Limnology, Chinese Academy of Sciences, Nanjing, PR China.
*Correspondence: Christopher A. Abin, cabin@ou.edu
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Abbreviations: AAI, Average amino acid identity; ANI, Average nucleotide identity; COGs, Clusters of Orthologous Groups; dDDH, Digital DNA-DNA hybridization; DeMMO, Deep Mine Microbial Observatory; DSMZ, Deutsche Sammlung von Mikroorganismen und Zellkulturen; GSIs, Gene Support Indices; ICMs, Intracytoplasmic membranes; PLFA, Phospholipid fatty acid; pMMO, Particulate methane monooxygenase; PQQ, Pyrroloquinoline quinone; sMMO, Soluble methane monooxygenase; SURF, Sanford Underground Research Facility.
The 16S rRNA genes of strains WSC-6^T, WSC-7^T, SURF-1^T, and SURF-2^T have been deposited at GenBank/EMBL/DDBJ under the accession numbers ON758928, ON758929, ON758926, and ON758927, respectively. The GenBank accession numbers for the draft genome sequences of strains WSC-6^T, WSC-7^T, SURF-1^T, and SURF-2^T are JANIBK000000000, JANIBL000000000, JANIBM000000000, and JANIBJ000000000, respectively.
Thirteen supplementary figures and four supplementary tables are available with the online version of this article

'*M. rhizoryzae*' [12], and '*M. rubra*' [13] have been described, but not validly named. Unfortunately, type strains no longer exist for *M. aurantiaca*, *M. fodinarum*, and *M. scandinavica* [14].

Members of the genus *Methylomonas* are Gram-negative, rod-shaped or coccoid, neutrophilic, mesophilic, and may produce pink, red, or orange carotenoid pigments [14]. Recognized species can only use methane or methanol as the sole sources of carbon and energy. Cells possess type I intracytoplasmic membranes which presumably harbour the methane-oxidizing enzymatic machinery. Representatives of this genus have been isolated from a variety of non-saline, non-thermal environments, including a freshwater aquatic plant [1], sewage sludge and wastewater [3, 4], coal mine drainage [3], groundwater [9], rice paddy [5], peat bog lake [7], manure pit [6], river water [2], and lake sediments [8].

Here, we report the characterization of four novel representatives of the genus *Methylomonas* from a freshwater creek in southwestern Oklahoma and the deep terrestrial subsurface at the Sanford Underground Research Facility (SURF) in South Dakota. We propose to assign these isolates to four new species within the genus, namely *Methylomonas rivi* sp. nov., *Methylomonas rosea* sp. nov., *Methylomonas aurea* sp. nov., and *Methylomonas subterranea* sp. nov.

ISOLATION AND ECOLOGY

Strains WSC-6^T and WSC-7^T were isolated from a microbial biofilm on a submerged rock in Saddle Mountain Creek, a tributary of the Washita River in Kiowa County, Oklahoma, USA (GPS coordinates 35.002556 N, 98.688028 W). The biofilm was situated in a zone where creek water mixed with anoxic, sulphur- and sulphide-rich artesian fluids emanating from nearby Zodletone Spring (Fig. S1, available in the online version of this article) [15–18]. A combination of light exposure, elevated sulphide concentrations, and semi-oxic conditions at the site has led to the development of extensive filamentous, sulphide-oxidizing microbial communities. Approximately 50 ml biofilm material was collected in June 2018 using sterile tongs and kept on ice until processed at our laboratory at the University of Oklahoma.

Strains SURF-1^T and SURF-2^T were isolated from a microbial mat and submerged sediments, respectively, inside a tunnel at SURF in Lead, South Dakota, USA (Fig. S2). SURF is hosted within the former Homestake gold mine and houses active projects in physics, engineering, geology, and biology, including the Deep Mine Microbial Observatory (DeMMO) [19, 20]. The tunnel was located within the '17 Ledge' mining area approximately 4850 ft (1478 m) below the surface. The floor of the tunnel was submerged beneath ~2–3 cm of standing water fed by flows from fractures in the walls and ceiling. These fluids have an average temperature of 31–33 °C and a pH of 8.1–9.1 [19, 20]. White- and orange-coloured microbial mats had become established at several of the fractures. We used sterile tongs and spatulas to retrieve approximately 50 ml mat and sediment samples in September of 2018 from a location ~90 m down tunnel from the borehole at DeMMO site D5 (formerly borehole no. 11938). Samples were stored in sterile 50 ml conical tubes and immediately placed on ice for shipment back to the laboratory.

Microcosms from each site were established using 5 g wet biomass in sealed, 160 ml glass serum bottles under a headspace of air amended with 1% methane. Microcosms were incubated statically in the dark at room temperature. Methane depletion in the microcosms was monitored using a Shimadzu GC-14A gas chromatograph with a flame ionization detector. Microcosms that actively depleted methane were serially diluted tenfold in 20 ml of HPB-1 (strains WSC-6^T and WSC-7^T) or HPB-2 (strains SURF-1^T and SURF-2^T) media (see Table S1 for media composition) and incubated in sealed, 160 ml glass serum bottles under a headspace of air amended with 1% methane. Cultures that actively depleted methane were again serially diluted tenfold in HPB-1 or HPB-2 and plated onto 1.5% (w/v) HPB-1 or HPB-2 agar. Petri plates were incubated at room temperature for 3–4 weeks in gas-tight ammunition boxes under an atmosphere of air amended with 4% methane. Colonies that grew were picked and purified by successive rounds of dilution streaking on 1.5% (w/v) HPB-1 or HPB-2 agar. Pure cultures were transferred to liquid HPB-1 or HPB-2 in sealed glass serum bottles and tested for the ability to deplete methane in the headspace. Methane-depleting isolates were maintained in liquid culture in sealed glass serum bottles under a headspace of air amended with 4–10% methane. The novel strains were deposited in the American Type Culture Collection (ATCC) and the Leibniz-Institut – Deutsche Sammlung von Mikroorganismen und Zellkulturen (DSMZ) under the following accession numbers: WSC-6^T (=ATCC TSD-251^T=DSM 112293^T); WSC-7^T (ATCC TSD-252^T=DSM 112281^T); SURF-1^T (ATCC TSD-253^T=DSM 112282^T); and SURF-2^T (ATCC TSD-254^T=DSM 112283^T).

16S rRNA GENE PHYLOGENY

Genomic DNA from each strain was extracted using the Qiagen DNeasy Blood and Tissue Kit according to the manufacturer's instructions. The 16S rRNA genes were amplified using universal bacterial primers 27F (5′-AGAGTTTGATCACTGGCTCAG-3′) and 1492R (5′-ACGGTACCTTGTTACGACTT-3′) [21, 22]. The amplicons were sequenced from both ends using an Applied Biosystems 3130xl Genetic Analyzer with a Big Dye Terminator version 3.1 Cycle Sequencing Kit (Biology Core Molecular Laboratory, University of Oklahoma, Norman, OK, USA). The forward and reverse sequences were then assembled into nearly full-length 16S rRNA genes using Geneious version 11.1.5 (Biomatters Ltd.). Sequence similarities to closely related strains were calculated using the EzBioCloud identification tool (www.ezbiocloud.net/identify) [23] and BLAST [24] searches against the

NCBI GenBank database. The 16S rRNA sequences from each strain and closely related strains were aligned using MUSCLE [25]. Phylogenetic trees were reconstructed using the neighbour-joining [26], maximum likelihood [27], and maximum parsimony [28] methods with MEGA version 11.0.11 [29]. Evolutionary distances were computed using the Kimura two-parameter model [30] for the neighbour-joining and maximum-likelihood trees, whereas the subtree-pruning-regrafting algorithm was used for the maximum-parsimony trees. In all cases, support values for branches were calculated using the bootstrap resampling method [31] with 1000 replications.

Analysis of the 16S rRNA genes from each of the strains revealed that they were all most closely related to members of the genus *Methylomonas* in the family *Methylococcaceae* of the class *Gammaproteobacteria* (Fig. S3). Strains WSC-6^T and SURF-2^T exhibited the highest (98.6%) 16S rRNA gene sequence similarity to each other. The second-highest matches for strains WSC-6^T and SURF-2^T were *M. paludis* MG30^T (97.9%) and *M. fluvii* EbB^T (97.5%), respectively. Strains WSC-7^T and SURF-1^T were most closely related to *M. fluvii* EbB^T (99.6%) and *M. koyamae* Fw12E-Y^T (99.2%), respectively. Strain WSC-6^T clustered with *M. paludis* MG30^T, *M. scandinavica* SR5^T, and '*M. rubra*' 15sh in all three phylogenetic trees, albeit with low bootstrap support (Figs 1, S4 and S5). This clade also included *M. subterranea* SURF-2^T in the maximum-likelihood and neighbour-joining trees, but not the maximum-parsimony tree. Strain WSC-7^T consistently formed a clade with at least *M. fluvii* EbB^T, *M. defluvii* OY6^T, *M. methanica* NCIMB 11130^T, and '*M. denitrificans*' FJG1 regardless of the algorithm used to reconstruct the phylogenetic trees. Similarly, strain SURF-1^T formed a strongly supported clade with *M. koyamae* Fw12E-Y^T and both strains of *M. rapida* in all three phylogenetic trees.

GENOME FEATURES

Genomic DNA from each of the novel strains was sheared ultrasonically using a QSonica 800R3 sonicator. Illumina-compatible shotgun libraries were constructed using a Roche KAPA HyperPrep Kit according to the manufacturer's instructions. The paired-end libraries were sequenced on an Illumina MiSeq instrument using a MiSeq Reagent Kit version 2 (2×250 bp paired-end; strains WSC-6^T and WSC-7^T) or version 3 (2×300 bp paired-end; strains SURF-1^T and SURF-2^T). Processing of the raw sequence data to remove reads with at least one uncalled base ('N'), trim low-quality (Q<30) bases from both ends, and merge overlapping reads was performed using fastp version 0.23.2 [32]. *De novo* assembly of the analysis-ready reads was performed using SKESA version 2.5.1 [33]. The quality of the draft genome assemblies, with respect to completeness and contamination, was evaluated using CheckM version 1.2.2 [34]. The genomes were annotated using the NCBI Prokaryotic Genome Annotation Pipeline version 6.5 [35–37]. Putative protein-coding genes were assigned KEGG Orthology (KO) [38] numbers using BlastKOALA [39]. Metabolic pathways were reconstructed from the KO numbers using the KEGG Mapper Reconstruct Tool [40].

For phylogenomic and pangenomic analyses, reference genomes for representative members of the family *Methylococcaceae* were obtained from the NCBI RefSeq database. Pairwise average nucleotide identity (ANI) and average amino acid identity (AAI) scores were calculated using FastANI version 1.33 [41] and ezAAI version 1.2 [42], respectively. Digital DNA–DNA hybridization (dDDH) values were obtained using formula 2 of the Genome-to-Genome Distance Calculator version 3.0 [43, 44]. Sequences for 92 single-copy core bacterial genes were retrieved from each genome using the Up-to-Date Bacterial Core Gene (UBCG) version 3.0 pipeline [45]. The sequences were aligned using MUSCLE, filtered to remove invariant sites, and concatenated. A maximum-likelihood amino acid tree was then reconstructed from the set of concatenated sequences using FastTree version 2.1.11 [46]. Evolutionary distance matrices were estimated using the Jones–Taylor–Thornton [47] + CAT [48] models with 1000 re-samplings. Unique gene clusters in the genomes of the novel isolates were identified by building a pangenome of the genus *Methylomonas* using Anvi'o version 8.0 [49] and annotated into Clusters of Orthologous Groups (COGs) functional categories [50, 51]. Lastly, full length *pmoA* gene sequences from each of the novel strains and related members of the family *Methylococcacea* were aligned using MUSCLE. Phylogenetic trees were reconstructed in MEGA version 11.0.11 using the maximum-likelihood, neighbour-joining, and maximum-parsimony methods with 1000 bootstrap replications.

Detailed information about the genome assemblies is provided in Table S2. A total of 2071760–8593614 quality-filtered reads were used to generate each draft genome, which ranged in size from 4.41 to 4.81 Mbp. These values fall within the range of 3.67–5.59 Mbp observed for reference genomes from other members of the genus *Methylomonas*. Estimates of genome coverage, completeness, and contamination ranged from 71× to 295×, from 92.1 to 95.8%, and from 1.2 to 5.4%, respectively. The DNA G+C content ranged from 51.5 to 56.0 mol%. Pairwise comparisons of genomic similarity (ANI, AAI, and dDDH) are presented in Figs S6–S8. Strains WSC-6^T and SURF-2^T were more closely related to each other (84.5% ANI, 85.8% AAI, and 27.4% dDDH) than to any other described member of the genus *Methylomonas* for which a reference genome existed. Strains WSC-7^T and SURF-1^T exhibited the highest similarity to *M. methanica* NCIMB 11130^T (88.4% ANI, 92.4% AAI, and 35.0% dDDH) and *M. koyamae* Fw12E-Y^T (88.3% ANI, 91.3% AAI, and 34.3% dDDH), respectively. All of these values fall below the currently accepted thresholds of 95–96% ANI [52, 53], 95% AAI [54], and 70% dDDH [55] for the delineation of microbial species boundaries, indicating that all four strains represent novel species of the genus

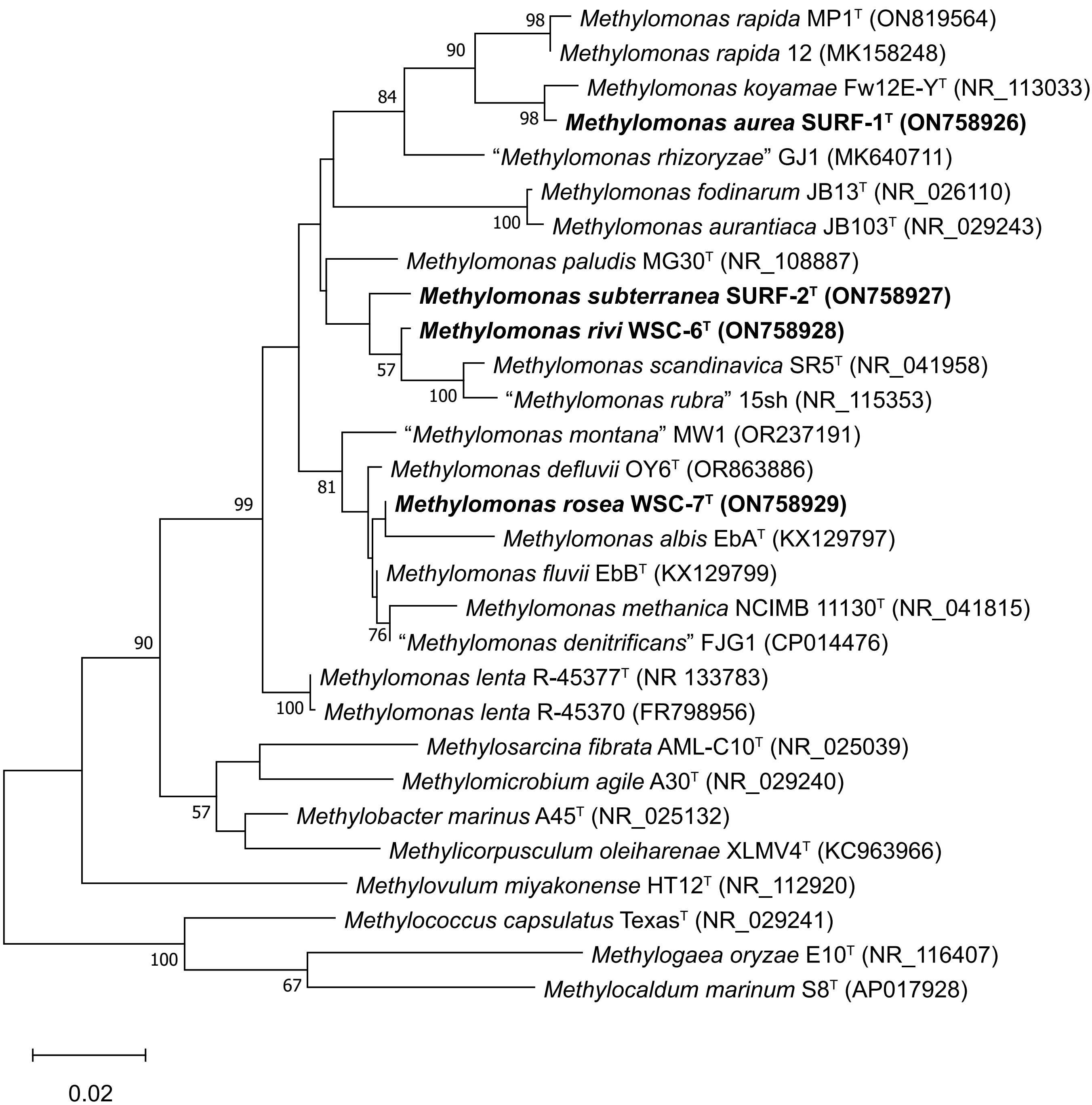

Fig. 1. Maximum likelihood phylogenetic tree based on partial 16S rRNA gene sequences (1385 positions) showing the relationship between the novel isolates and other taxonomically characterized members of the family *Methylococcaceae*. Bootstrap values are expressed as percentages of 1000 replications. Only those values ≥50% are shown at branch nodes. GenBank accession numbers are given in parentheses. Bar, 0.02 substitutions per nucleotide position.

Methylomonas. This conclusion was supported by a genome-based phylogeny, which mirrored the 16S rRNA gene-based phylogenies in showing that the strains formed distinct lineages within the genus (Fig. 2).

A pangenome analysis of the genus *Methylomonas* revealed that the core genome comprised 1711 gene clusters, of which 1488 consisted of single-copy genes (Fig. S9). The number of unique gene clusters in the draft genomes of each of the novel strains ranged from 319 to 645, which is comparable to the range (23–889) found in other taxonomically characterized

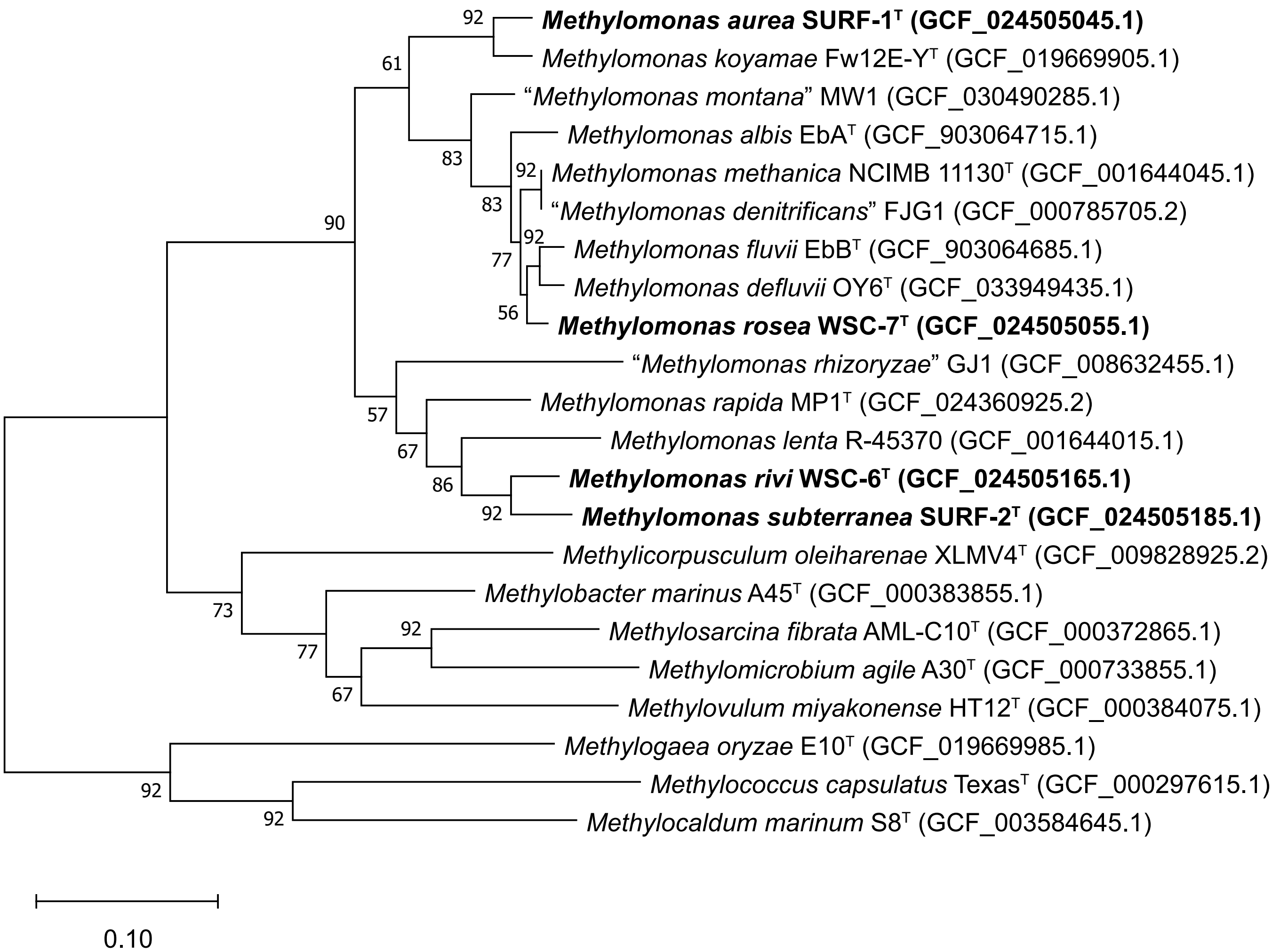

Fig. 2. Maximum likelihood phylogenomic tree, based on the comparative analysis of concatenated amino acid sequences from 92 single-copy core proteins (29565 positions), showing the position of the novel isolates in relation to other taxonomically characterized members of the family *Methylococcaceae*. Gene Support Indices (GSIs), which indicate how many genes support any given branch, are provided at the branch nodes. Only those GSI values ≥50% are shown. RefSeq accession numbers are given in parentheses. Bar, 0.10 substitutions per amino acid position.

members of the genus (Table S3). Between 56.5–73.3% unique genes from each of the novel strains could not be assigned to a COG functional category. The remaining genes were found to be associated with cell envelope biosynthesis (11.6–21.5%), antiviral defence (6.3–15.9%), and signal transduction (8.0–11.6%) (Fig. S10). Strain WSC-6^T also harboured unique genes involved in the transport and metabolism of secondary metabolites (6.0%) and inorganic ions (9.9%). Genes associated with transcription and mobile elements accounted for 8.9 and 10.7%, respectively, of those unique to strain SURF-1^T, while others involved in replication, recombination and repair comprised 10.5% of the annotated genes unique to strain WSC-7^T.

Key genes involved in methane oxidation were found in the genomes of all four strains (Table S4). Each contained a single copy of the *pmoCAB* operon encoding three subunits of the canonical, copper-containing particulate methane monooxygenase (pMMO) [56]. The *pmoA* genes from the novel strains clustered together with those from other members of the genus *Methylomonas* in phylogenetic trees reconstructed using three different methods (Figs 3, S11, and S12). The *pmoA* genes from strains WSC-6^T and SURF-2^T exhibited the highest similarity (96.5%) to each other, while those from strains WSC-7^T and SURF-1^T were most similar to *M. defluvii* OY6^T (97.1%) and '*M. montana*' MW1 (94.0%), respectively. Also present in each genome was a single copy of the *pxmABC* operon, which encodes a divergent particulate methane monooxygenase (pXMO) produced by some gammaproteobacterial methanotrophs [57]. While the exact function of this enzyme is not yet fully known, it is thought to contribute towards methane oxidation under hypoxic conditions [10, 58]. The genome of strain WSC-7^T was the only one found to harbour a copy of the *mmoXYBZDC* operon, which encodes a soluble methane monooxygenase (sMMO) in addition to an NADH-dependent

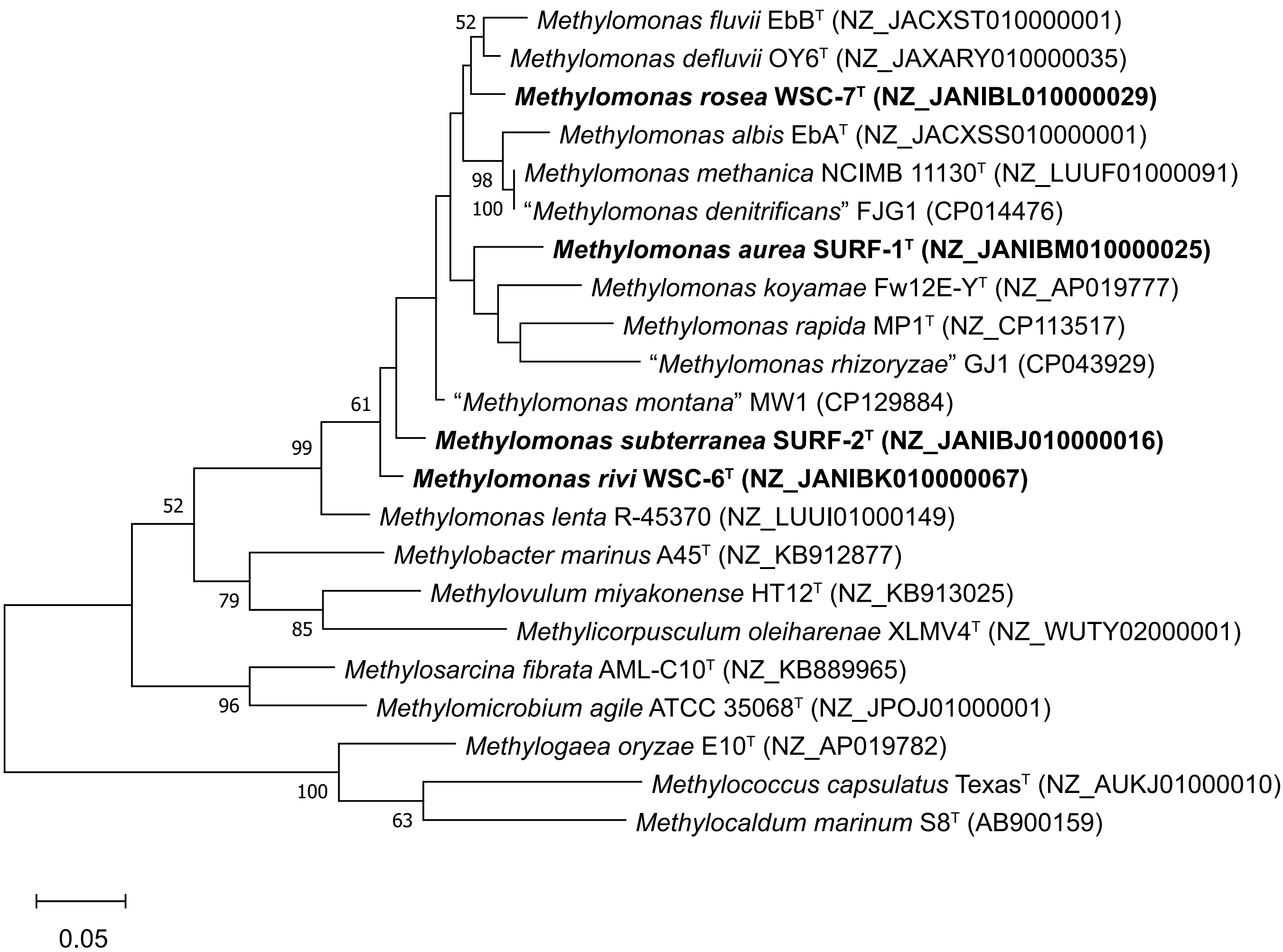

Fig. 3. Maximum likelihood phylogenetic tree based on full length *pmoA* gene sequences (750 positions), showing the position of the novel strains in relation to other taxonomically characterized members of the family *Methylococcaceae*. Bootstrap values are expressed as percentages of 1000 re-samplings. Only those values ≥50% are shown at the branch nodes. Genbank or RefSeq accession numbers are given in parentheses. Bar, 0.05 substitutions per nucleotide position.

reductase and several regulatory proteins. Unlike the copper-containing and membrane-integral pMMO and pXMO, this enzyme complex is iron-dependent and localized in the cytosol [56].

Aerobic methanotrophs oxidize methanol to formaldehyde using two different types of methanol dehydrogenases. Gene clusters encoding the pyrroloquinoline quinone (PQQ)-linked, calcium-dependent (*mxaFJGIRPSACKL*) [59] and lanthanide-dependent (*xoxFJ*) [60] methanol dehydrogenases as well as a pathway for PQQ biosynthesis (*pqqABCDE*) were at least partially found in the genomes of each strain. All four strains appear to oxidize formaldehyde to formate via the tetrahydromethanopterin (H_4MPT)-dependent pathway [61] and do not possess a dedicated formaldehyde dehydrogenase. The gene clusters (*fae*, *mtdB*, *mch*, and *fhcBADC*) responsible for encoding enzymes of this pathway were detected in each strain. Lastly, each strain possessed a single copy of the *fdsGBACD* operon, which encodes a soluble, NAD^+-dependent formate dehydrogenase [62].

Some gammaproteobacterial methanotrophs have been shown to perform denitrification or nitrogen fixation under hypoxic conditions. For example, '*M. denitrificans*' FJG1 and *Methylomicrobium album* BG8^T can couple methane oxidation to the dissimilatory reduction of nitrate and nitrite, respectively, at dissolved oxygen concentrations <50 nM [10, 58]. In both cases, denitrification was incomplete and the end product was nitrous oxide. The genome of '*M. denitrificans*' FGJ1 encodes several enzymes of the denitrification pathway, including two nitrate reductases (*narGHJI* and *napABC*), two nitrite reductases (*nirK* and *nirS*), and nitric oxide reductase (*norCB*) [10]. A search for similar genes in the genomes of the novel strains revealed that they each harboured a single copy of the *norCB* operon (Table S4). No other genes involved unequivocally in the dissimilatory reduction of nitrogen oxides were found. Additionally, all four strains possessed the *nifHDK* gene cluster encoding a molybdenum-iron (Mo-Fe)-type nitrogenase responsible for catalysing the six-electron reduction of dinitrogen to ammonia [63]. The Mo-Fe nitrogenase gene cluster has been found in the genomes of many alpha- and gammaproteobacterial methanotrophs, including several members of the genus *Methylomonas* [64].

PHYSIOLOGY AND CHEMOTAXONOMY

Colony morphology and colour were evaluated for cultures grown on HPB-1 and HPB-2 agar. Cell morphology and motility were analysed using an Olympus BX41 phase contrast microscope equipped with a Jenoptik Gryphax Subra camera. The imaged cells had been grown to mid-exponential phase in liquid culture using methane as the sole source of carbon and energy. Cell size measurements were made using Jenoptik Gryphax imaging software version 2.0.0.66. At least ten randomly selected cells per field were used to calculate an average cellular length and width for each strain. Cells for transmission electron microscopy were harvested from mid-exponential phase cultures and prepared as described previously [65]. Thin sections were examined at 80 kV on a JEOL 2010F transmission electron microscope equipped with a cold field emission electron source and a Direct Electron DE-12 camera. The Gram stain was performed according to a modification of Hucker's method [66]. The production of cysts or exospores by cultures grown to stationary phase was assessed using phase contrast microscopy. The catalase test was performed by adding cells to a drop of 3% H_2O_2 and observing for the formation of gas bubbles. Cytochrome oxidase activity was tested with Bioanalyse oxidase strips.

Heat and desiccation resistance tests were performed on dense (~10^{11} cells ml^{-1}) cell suspensions harvested from late-exponential phase batch cultures. For the heat resistance test, cell suspensions were incubated in a water bath at 80 °C for 10 min, cooled on ice for 5 min, and then plated onto 1.5% (w/v) HPB-1 or HPB-2 agar. The Petri plates were then incubated at 25 °C for 3 weeks inside sealed ammunition boxes under an atmosphere containing 4% methane and 5% oxygen. For the desiccation resistance test, drops of cell suspensions were air-dried inside sterile Petri plates for 3 weeks at 25 °C. The dried cells were then rehydrated with liquid HPB-1 or HPB-2, plated onto 1.5% (w/v) HPB-2 or HPB-2 agar, and incubated at 25 °C for 3 weeks.

The effects of temperature (4–50 °C), pH (5.0–10.1), and NaCl concentration (0–5% w/v) on the growth of each strain were tested in liquid HPB-1 or HPB-2 media by measuring the OD_{600} of cultures on a Thermo Scientific Spectronic 20D+spectrophotometer. The media were adjusted to each pH value by a combined buffer of 10 mM sodium citrate, 10 mM MOPS, 1 mM Na_2HPO_4, 10 mM Tris, and 10 mM glycine. The final pH of the media was confirmed following preparation. Tests for the utilization of carbon sources for growth were performed in liquid HPB-1 or HPB-2 without methane added to the headspace. Unless otherwise noted, the media were supplemented with 0.1% (w/v) of the following filter-sterilized carbon sources: methanol, formaldehyde (0.01%), formate, urea, methylamine, dimethylamine, dimethyl carbonate, formamide, acetate, pyruvate, succinate, malate, ethanol, citrate, glycine, D-xylose, maltose, and glucose. The range of nitrogen sources for growth was tested by replacing nitrate in HPB-1 or HPB-2 with 2 mM (unless otherwise noted) of the following compounds: nitrite, ammonium (as NH_4Cl), formamide, methylamine, glycine, hydroxylamine, urea, L-serine, L-proline, L-aspartate, L-leucine, L-arginine, and yeast extract (0.1% w/v). Growth on potential nitrogen sources was compared to nitrate-free controls. The ability to fix nitrogen was assessed by incubating the strains in nitrate-free HPB-1 or HPB-2 under a headspace containing 10% methane and either ambient (21%) or low (2%) oxygen, balanced with nitrogen gas. Carryover nitrate was reduced as much as possible by washing the cells three times with nitrate-free media prior to the start of the experiment. Nitrate-free cultures, flushed with helium to remove nitrogen gas from the headspace, and amended with 10% methane and 2% or 21% oxygen, served as controls for growth.

Cells for fatty acid analysis were grown in HPB-1 or HPB-2 media at room temperature under a headspace consisting of air amended with 10% methane. Approximately 30 mg wet biomass from each strain was harvested in the late exponential phase by centrifugation, freeze-dried, and shipped to the Identification Service of the DSMZ for analysis. Cellular fatty acids were converted into fatty acid methyl esters and analysed by gas chromatography with flame ionization detection using the MIDI Sherlock Microbial Identification System. Gas chromatography–mass spectrometry was used to resolve summed features and confirm the identity of the fatty acids.

The cells of all four strains were short, Gram-negative, motile rods approximately 1.2–2.1 µm long and 1.0–1.4 µm wide (Fig. 4). Cells occurred singly or as pairs of dividing cells during logarithmic growth. Transmission electron microscopy of thin sections of cells showed that all four strains produced intracytoplasmic membranes (ICMs) arranged as stacks of vesicular discs (Fig. 5). The orientation of the ICMs was characteristic of type I gammaproteobacterial methanotrophs. Colonies of strain WSC-6^T were flat, pale pink, and circular, with a diameter of 1–2 mm after 3 weeks of incubation at room temperature on HPB-1 agar Petri plates. Strain WSC-7^T displayed phenotypic variation when grown on solid media. Where distinct colonies occurred, they were small (0.75–1.0 mm in diameter), raised, circular, and pale pink in colour. Otherwise, petri plates often developed large, pink, irregular, slimy masses that could exceed 30–40 mm long. Strains SURF-1^T and SURF-2^T formed orange and pink circular colonies, respectively, on HPB-2 agar Petri plates. Colonies of strain SURF-1^T were 1–5 mm in diameter while those of SURF-2^T were 1–2 mm wide following 3 weeks of growth.

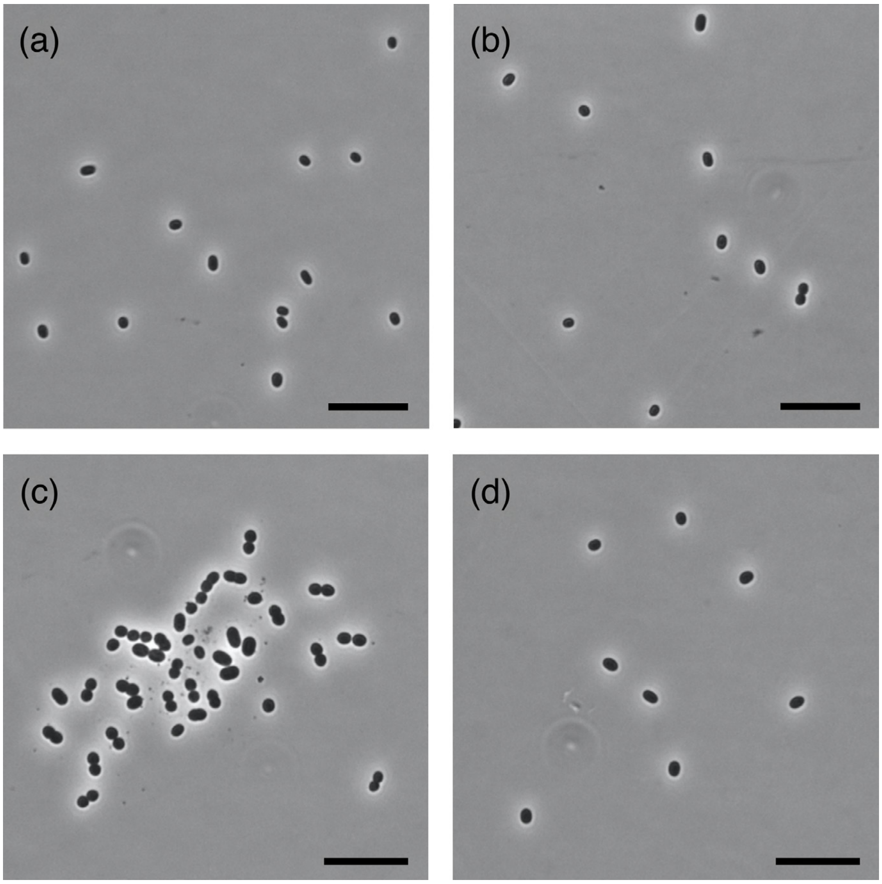

Fig. 4. Phase contrast microscope images of cells of strains (a) WSC-6^T, (b) WSC-7^T, (c) SURF-1^T, and (d) SURF-2^T. Bar, 10 µm.

When grown statically in liquid culture, cells of each of the strains were not evenly dispersed. Instead, they formed aggregates that were suspended or loosely attached to the bottom and sides of glass serum bottles. This phenomenon was most pronounced with cultures of strain WSC-7^T, which formed large, cohesive clumps of cells if left undisturbed for 3–4 weeks. Cells of strain SURF-1^T did not form films, but rather gold-coloured floc that settled onto the bottom of serum bottles. Liquid cultures of all four strains could be rendered homogenous with continuous shaking at 250 r.p.m. Continuous shaking of the cultures made it possible to track cell growth on a spectrophotometer.

All four strains were sensitive to heat and desiccation. Cells were not observed to produce cysts or exospores, regardless of growth phase. The strains remained viable in liquid culture at room temperature for ~1 month or on adequately hydrated Petri plates for ~1–2 months. For long-term storage, all four strains could be successfully preserved at −80 °C using 5% dimethyl sulphoxide (DMSO) as the cryoprotectant. The oxidase and catalase tests were performed on cells grown under low (2%) and high (21%) headspace oxygen concentrations to determine whether test results were affected by oxygen levels. All strains tested negative for oxidase and weakly positive for catalase under both conditions.

Strains WSC-6^T and WSC-7^T grew at temperatures of 4–35 °C, with optima at 25 and 20 °C, respectively. Strains SURF-1^T and SURF-2^T shared the same temperature optimum (40 °C) and range (10–45 °C) for growth. Strains WSC-6^T and WSC-7^T grew best at pH 7.5 (range, pH 6.5–9.3) and pH 8.8 (range, pH 6.2–9.9), respectively (Fig. S13). Strain SURF-1^T grew at pH 7.0–10.1, with an optimum of pH 8.8. This value is consistent with the range of pH (pH 8.1–9.1) recorded previously in the fracture fluids that flowed through the microbial mat from which the strain was isolated [19, 20]. Strain SURF-2^T grew best at pH 7.3 but had a narrow range for growth (pH 6.8–7.7). Strains WSC-6^T and WSC-7^T grew at salinities of 0–0.75% (optimum, 0.1%) and 0–0.2% (optimum, 0%) NaCl, respectively. Strains SURF-1^T and SURF-2^T grew at salinities of 0–0.5% (optimum, 0.1%) and 0–1.5% (optimum, 0.5%) NaCl, respectively.

Methane and methanol were the only compounds that could serve as carbon and energy sources for growth of each strain. Strains WSC-6^T, WSC-7^T, and SURF-2^T could grow at a maximum methanol concentration of 0.1% while strain SURF-1^T could only tolerate up to 0.05%. Growth of each strain on methanol through multiple transfers was unstable, a phenomenon that has

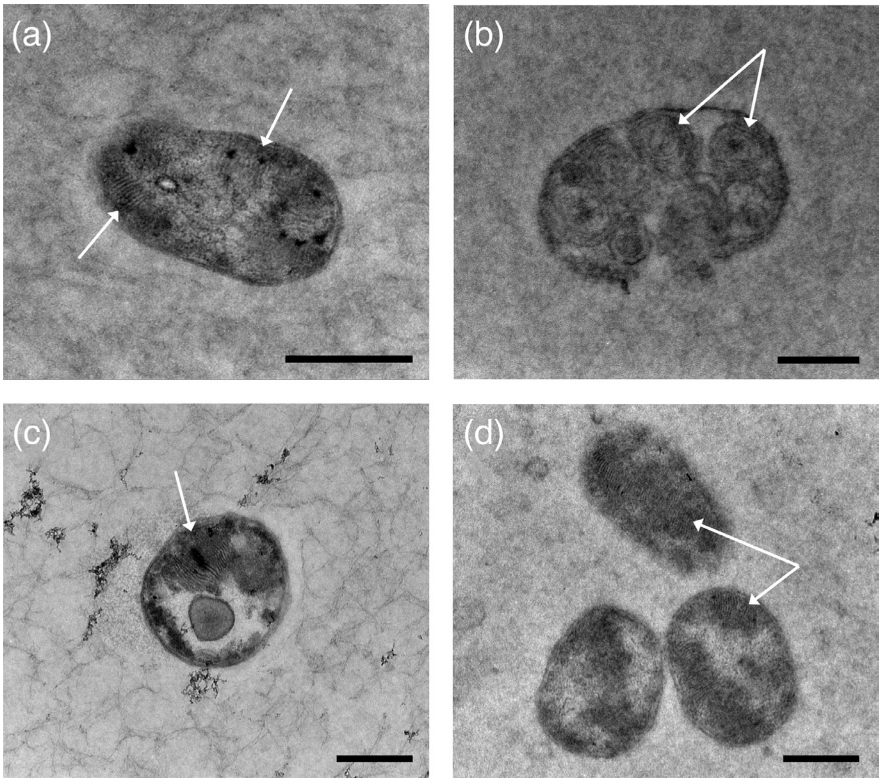

Fig. 5. Transmission electron microscope images of ultra-thin sections of cells of strains (a) WSC-6^T, (b) WSC-7^T, (c) SURF-1^T, and (d) SURF-2^T. Intracytoplasmic membranes are highlighted by arrows. Bar, 0.5 μm.

been reported previously for *M. rapida* [8]. Strains WSC-6^T and WSC-7^T grew using nitrate, nitrite, urea, L-cysteine, and yeast extract as sole nitrogen sources. Strain WSC-6^T could also use ammonium (as NH_4Cl) as a nitrogen source. Strains SURF-1^T and SURF-2^T grew with nitrate, L-cysteine, and yeast extract as nitrogen sources. L-Serine could also serve as a nitrogen source for strain SURF-1^T. All four strains were capable of nitrogen fixation, but only at low oxygen tension (2% headspace concentration).

All four strains had phospholipid fatty acid (PLFA) profiles characteristic of members of the genus *Methylomonas*, with a predominance of $C_{14:0}$, $C_{16:1}\,\omega 8c$, $C_{16:1}\,\omega 7c$, $C_{16:1}\,\omega 5c$, and $C_{16:0}$ fatty acids (Table 1). Strains WSC-6^T and SURF-1^T produced a large amount of the fatty acid $C_{16:1}\,\omega 7c$ (32.9 and 22.6%, respectively) in comparison to other members of the genus. Conversely, strains WSC-6^T and WSC-7^T produced a smaller amount of the fatty acid $C_{16:1}\,\omega 8c$ relative to all other members of the genus except *M. albis* EbA^T, *M. fluvii* EbB^T, and *M. defluvii* OY6^T, which were not found to produce it at all. The fatty acid $C_{16:1}\,\omega 5t$ made up 17.8% of the PLFA profile of strain WSC-6^T. This strain joins the type strains of *M. methanica*, *M. fodinarum, M. aurantiaca*, and *M. paludis* as the only members of the genus found to produce it. Strain WSC-6^T was also unique in that it was the only member of the genus to produce $C_{15:1}\,\omega 7c$, albeit in a low amount (1.0%). None of the strains produced appreciable amounts of the fatty acids $C_{12:0}$, $C_{13:1}$, $C_{13:0}$, $C_{15:1}\,\omega 6c$, or iso-$C_{15:1}$. These fatty acids made up a combined 37–38% of the PLFA profiles of *M. albis* EbA^T and *M. fluvii* EbB^T but are typically absent or present in very low quantities in all other members of the genus.

The phenotypic, phylogenomic, and chemotaxonomic data presented here suggest that each of the strains represent novel members of the genus *Methylomonas*. The major morphological, physiological, and genomic characteristics that distinguish the strains from other type strains of the genus are listed in Table 2. Strains WSC-6^T and SURF-2^T can be distinguished from each other based on their temperature optima for growth and salinity tolerances. Strain WSC-7^T and its closest neighbour, *M. methanica* NCIMB 11130^T, differ with respect to cyst and chain formation, pH optima for growth, oxidase test results, and tolerance of 1% NaCl. Strain SURF-1^T and *M. koyamae* Fw12E-Y^T differ with respect to cyst formation, pH optima for growth, oxidase test results, ability to tolerate 0.1% methanol, and ability to grow with even dispersion in liquid culture. Based on these differences, strains WSC-6^T, WSC-7^T, SURF-1^T, and SURF-2^T should be considered to represent four new species of the genus *Methylomonas*, for which we propose the names *Methylomonas rivi* sp. nov., *Methylomonas rosea* sp. nov., *Methylomonas aurea* sp. nov., and *Methylomonas subterranea* sp. nov., respectively.

Table 1. Cellular fatty acid composition of strains WSC-6^T, WSC-7^T, SURF-1^T, and SURF-2^T, in comparison to other type strains of the genus *Methylomonas*

Strains: 1, WSC-6^T; 2, WSC-7^T; 3, SURF-1^T; 4, SURF-2^T; 5, other type strains of the genus *Methylomonas*, including *M. albis* EbA^T [2], *M. fluvii* EbB^T [2], *M. lenta* R-45377^T [6], *M. paludis* MG30^T [7], *M. koyamae* Fw12E-Y^T [5], *M. methanica* NCIMB 11130^T [67], *M. fodinarum* JB13^T [67], *M. aurantiaca* JB103^T [67], and *M. defluvii* OY6^T [4]. No fatty acid data have been reported for *M. scandinavica* [9] and *M. rapida* [8]. Values are percentages of total fatty acids. Only fatty acids detected at ≥1% are shown. —, Not detected or <1%.

Fatty acid	1	2	3	4	5
$C_{12:0}$	—	—	—	—	0–26.6
$C_{13:1}$	—	1.0	—	—	0–4.8
$C_{13:0}$	—	—	—	—	0–5.1
$C_{14:0}$	14.8	21.1	17.2	20.2	6.4–51.6
$C_{15:1}$ *ω8c*	—	—	—	—	0–2.3
$C_{15:1}$ *ω7c*	1.0	—	—	—	—
$C_{15:1}$ *ω6c*	—	—	—	—	0–7.5
$C_{15:1}$ *ω5c*	—	—	—	—	0–1.1
iso-$C_{15:1}$	—	—	—	—	0–4.3
iso-$C_{15:0}$	—	—	—	—	0–2.5
anteiso-$C_{15:0}$	—	—	—	—	0–2.4
$C_{15:0}$	—	1.4	—	—	0–5.8
$C_{15:0}$ 2-OH	—	—	—	—	0–1.4
$C_{16:1}$ *ω8c*	13.7	12.6	28.3	32.7	0–42.4
$C_{16:1}$ *ω7c*	32.9	13.9	22.6	16.3	0–15.3
$C_{16:1}$ *ω6c*	5.4	3.1	6.0	5.1	0–13.3
$C_{16:1}$ *ω5c*	2.3	21.6	18.9	17.9	0–19.3
$C_{16:1}$ *ω5t*	17.8	—	—	—	0–34.8
$C_{16:0}$	7.3	16.2	3.5	4.3	3.5–8.7
$C_{16:0}$ 3-OH	1.9	7.0	1.3	1.1	0–9.0
cyclo-$C_{17:0}$	—	—	—	—	0–2.1
$C_{18:1}$ *ω7c*	—	—	—	—	0–2.5
$C_{18:1}$ *ω5c*	—	—	—	—	0–1.7
$C_{18:0}$	—	—	—	—	0–1.2

DESCRIPTION OF *METHYLOMONAS RIVI* SP. NOV.

Methylomonas rivi (ri'vi. L. gen. n. *rivi*, of a rivulet or creek, referring to the environment from which the type strain was isolated).

Cells are Gram-negative, motile rods (1.3–2.1 µm long and 1.1–1.4 µm wide) that occur singly or in pairs and produce intracytoplasmic membranes characteristic of type I gammaproteobacterial methanotrophs. No cysts or spores are produced. Cells are oxidase-negative and weakly catalase-positive. Colonies are flat, circular, and pale pink in colour. Cultures grow optimally at 25 °C (range, 4–35 °C), pH 7.5 (range, pH 6.5–9.3), and salinity of 0.1% NaCl (range, 0–0.75%). Grows using methane or methanol as the only carbon and energy sources, but is sensitive to methanol concentrations above 0.1%. Grows with nitrate, nitrite, ammonium (as NH_4Cl), urea, L-cysteine, and yeast extract as sole nitrogen sources. Capable of nitrogen fixation, but only at low oxygen tension. The major cellular fatty acids (≥10% of fatty acid profile) are $C_{16:1}$ *ω7c*, $C_{16:1}$ *ω5t*, $C_{14:0}$, and $C_{16:1}$ *ω8c*. The DNA G+C content is 53.5 mol%.

The type strain, WSC-6^T (=ATCC TSD-251^T=DSM 112293^T), was isolated from a biofilm on a submerged rock at the confluence of Zodletone Spring and Saddle Mountain Creek waters in Kiowa County, Oklahoma, USA. The GenBank/EMBL/DDBJ/PIR accession numbers for the 16S rRNA gene and draft genome sequences of strain WSC-6^T are ON758928 and JANIBK000000000, respectively.

Table 2. Major characteristics that distinguish strains WSC-6^T, WSC-7^T, SURF-1^T, and SURF-2^T from other type strains of the genus *Methylomonas*

Strains: 1, WSC-6^T; 2, WSC-7^T; 3, SURF-1^T; 4, SURF-2^T; 5, *Methylomonas albis* EbA^T [2]; 6, *Methylomonas fluvii* EbB^T [2]; 7, *Methylomonas lenta* R-45377^T [6]; 8, *Methylomonas paludis* MG30^T [7]; 9, *Methylomonas koyamae* Fw12E-Y^T [5]; 10, *Methylomonas scandinavica* SR5^T [9]; 11, *Methylomonas methanica* NCIMB 11130^T [67, 68]; 12, *Methylomonas fodinarum* JB13^T [3]; 13, *Methylomonas aurantiaca* JB103^T [3]; 14, *Methylomonas rapida* MP1^T [8]; 15, *Methylomonas defluvii* OY6^T [4]. +, Positive; w, weak positive; –, negative; v, variable; NR, not reported.

Characteristic	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Cell shape	Short rods	Short rods	Short rods	Short rods	Rods	Rods	Rods	Rods	Rods	Rod-like ovoids	Rods	Rods	Rods	Short rods	Rods
Cell length (µm)	1.3–2.1	1.2–2.1	1.2–2.1	1.4–1.9	2.5	2.5	1.3–2.0	1.0–4.0	1.2–2.5	1.5–1.7	0.5–3.0	0.5–3.0	0.5–3.0	2.1	1.4–2.0
Cell width (µm)	1.1–1.4	1.0–1.4	1.0–1.4	1.0–1.4	0.8	0.8	0.6–0.9	1.0–1.5	0.8–1.1	0.6–0.8	0.5–1.0	0.5–1.0	0.5–1.0	1.1	0.7–0.9
Colony pigmentation	Pale pink	Pale pink to pink	Orange	Pink	Pink	Pink	White to pink	Pale pink	Pink to orange	Pink	Pink	Orange	Orange	Dark Orange to Red	Pink
Motility	+	+	+	+	–	–	+	–	+	+	+	+	+	+	+
Cyst formation	–	–	–	–	–	–	–	–	+	NR	+	+	+	NR	NR
Chain formation	–	–	–	–	NR	NR	–	+	–	+	v	–	–	–	–
Growth with even dispersion	–	–	–	–	NR	NR	NR	+	+	NR	–	–	–	+	+
Temperature range for growth (°C)	4–35	4–35	10–45	10–45	1–30	1–30	15–28	8–30	10–40	5–30	10–35	10–40	10–30	8–45	20–37
Temperature optimum for growth (°C)	25	20	40	40	15	15	20	20–25	30	15	25–30	30–35	20–25	35	30
pH range for growth	6.5–9.3	6.2–9.9	7.0–10.1	6.8–7.7	6.0–9.0	5.5–9.0	6.3–7.8	3.8–7.3	5.5–7.0	5.0–9.5	5.5–9.0	5.0–9.0	5.5–9.0	5.5–7.3	4.5–7.5
pH optimum for growth	7.5	8.8	8.8	7.3	6.0	6.0	6.8–7.3	5.8–6.4	6.5	6.8–7.6	7.0	7.0	7.0	6.6–7.2	6.5
Tolerance of 1% NaCl	–	–	–	+	–	+	+	–	–	+	+	+	+	–	–
Growth with 0.1% methanol	+	+	–	+	–	+	–	+	+	+	+	+	+	+	+
Catalase	w	w	w	w	NR	NR	–	NR	+	NR	+	+	+	NR	+
Oxidase	–	–	–	–	NR	NR	+	NR	+	NR	+	–	–	NR	–
Nitrogen fixation	+	+	+	+	w	–	–	+	+	NR	v	+	+	–	+
sMMO (genes or activity)	–	+	–	–	+	+	–	–	–	NR	v	NR	NR	–	+
DNA G+C content (mol%)	53.5	51.5	56.0	54.5	51.1	51.5	47.0	48.5	57.1	53.8	52.1	58.4	56.5	52.8	51.7
Isolation source	Creek	Creek	Mine biofilm	Mine sediments	River water	River water	Manure pit	Peat bog lake	Rice paddy	Igneous rock groundwater	Freshwater sediments	Coal mine drainage	Sewage sludge	Lake sediments	Sewage sludge

DESCRIPTION OF *METHYLOMONAS ROSEA* SP. NOV.

Methylomonas rosea (ro'se.a. L. fem. adj. *rosea*, rose-coloured or pink, referring to the pink colour of colonies).

Cells are Gram-negative, motile rods (1.2–2.1 µm long and 1.0–1.4 µm wide) that occur singly or in pairs and produce intracytoplasmic membranes characteristic of type I gammaproteobacterial methanotrophs. No cysts or spores are produced. Cells are oxidase-negative and weakly catalase-positive. Colonies are raised, circular, and pale pink in colour. Distinct colonies co-occur with larger slimy, irregular, pink masses of cells. Cultures grow optimally at 20 °C (range, 4–35 °C), pH 8.8 (range, pH 6.2–9.9), and salinity of 0% NaCl (range, 0–0.2%). Grows using methane or methanol as the only carbon and energy sources, but is sensitive to methanol concentrations above 0.1%. Grows with nitrate, nitrite, urea, L-cysteine, and yeast extract as the sole nitrogen sources. Capable of nitrogen fixation, but only at low oxygen tension. The major cellular fatty acids (≥10% of fatty acid profile) are $C_{16:1}$ $\omega 5c$, $C_{14:0}$, $C_{16:0}$, $C_{16:1}$ $\omega 7c$, and $C_{16:1}$ $\omega 8c$. The DNA G+C content is 51.5 mol%.

The type strain, WSC-7^T (=ATCC TSD-252^T=DSM 112281^T), was isolated from a biofilm on a submerged rock at the confluence of Zodletone Spring and Saddle Mountain Creek waters in Kiowa County, Oklahoma, USA. The GenBank/EMBL/DDBJ/PIR accession numbers for the 16S rRNA gene and draft genome sequences of strain WSC-7^T are ON758929 and JANIBL000000000, respectively.

DESCRIPTION OF *METHYLOMONAS AUREA* SP. NOV.

Methylomonas aurea (au're.a. L. fem. adj. *aurea*, golden, referring to the gold colour of liquid cultures).

Cells are Gram-negative, motile rods (1.2–2.1 µm long and 1.0–1.4 µm wide) that occur singly or in pairs and produce intracytoplasmic membranes characteristic of type I gammaproteobacterial methanotrophs. No cysts or spores are produced. Cells are oxidase-negative and weakly catalase-positive. Colonies are raised to convex, circular, and orange in colour. Cultures grow optimally at 40 °C (range, 10–45 °C), pH 8.8 (range, pH 7.0–10.1), and salinity of 0.1% NaCl (range, 0–0.5%). Grows using methane or methanol as the only carbon and energy sources, but is sensitive to methanol concentrations above 0.05%. Grows with nitrate, L-serine, L-cysteine, and yeast extract as sole sources of nitrogen. Capable of nitrogen fixation, but only at low oxygen tension. The major cellular fatty acids (≥10% of fatty acid profile) are $C_{16:1}$ $\omega 8c$, $C_{16:1}$ $\omega 7c$, $C_{16:1}$ $\omega 5c$, and $C_{14:0}$. The DNA G+C content is 56.0 mol%.

The type strain, SURF-1^T (ATCC TSD-253^T=DSM 112282^T), was isolated from a microbial mat growing at a fracture approximately 4850 ft (1478 m) below the surface at the Sanford Underground Research Facility in Lead, South Dakota, USA. The GenBank/EMBL/DDBJ/PIR accession numbers for the 16S rRNA gene and draft genome sequences of strain SURF-1^T are ON758926 and JANIBM000000000, respectively.

DESCRIPTION OF *METHYLOMONAS SUBTERRANEA* SP. NOV.

Methylomonas subterranea (sub.ter.ra'ne.a. L. fem. adj. *subterranea*, underground, below the Earth's surface, referring to the site from which the strain was isolated).

Cells are Gram-negative, motile rods (1.4–1.9 µm long and 1.0–1.4 µm wide) that occur singly or in pairs and produce intracytoplasmic membranes characteristic of type I gammaproteobacterial methanotrophs. No cysts or spores are produced. Cells are oxidase-negative and weakly catalase-positive. Colonies are convex, circular, and pink in colour. Cultures grow optimally at 40 °C (range, 10–45 °C), pH 7.3 (range, pH 6.8–7.7), and salinity of 0.5% NaCl (range, 0–1.5%). Grows using methane or methanol as the only carbon and energy sources, but is sensitive to methanol concentrations above 0.1%. Grows with nitrate, L-cysteine, and yeast extract as sole sources of nitrogen. Capable of nitrogen fixation, but only at low oxygen tension. The major cellular fatty acids (≥10% of fatty acid profile) are $C_{16:1}$ $\omega 8c$, $C_{14:0}$, $C_{16:1}$ $\omega 5c$, and $C_{16:1}$ $\omega 7c$. The DNA G+C content is 54.5 mol%.

The type strain, SURF-2^T (ATCC TSD-254^T=DSM 112283^T), was isolated from submerged sediments in a tunnel approximately 4850 ft (1478 m) below the surface at the Sanford Underground Research Facility in Lead, South Dakota, USA. The GenBank/EMBL/DDBJ/PIR accession numbers for the 16S rRNA gene and draft genome sequences of strain SURF-2^T are ON758927 and JANIBJ000000000, respectively.

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Author contributions

Conceptualization, C.A.A., K.S., and L.R.K.; methodology, C.A.A., K.S., P.A.L., and L.R.K.; resources, K.S., P.A.L., and L.R.K.; investigation, C.A.A., C.T.G., R.A.S., K.F.K., R.M.G., S.C.B., and H.C.; data curation, C.A.A., C.T.G., H.C., and K.S.; formal analysis, C.A.A., C.T.G., H.C., K.S, and L.R.K.; visualization, C.A.A.; validation, C.A.A., C.T.G., K.S., P.A.L., and L.R.K.; writing – original draft, C.A.A.; writing – review and editing, C.A.A., C.T.G., K.S., R.A.S., K.F.K., R.M.G., S.C.B., H.C., P.A.L., and L.R.K.; project administration, K.S. and L.R.K.; supervision, K.S., P.A.L., and L.R.K.; funding acquisition, K.S. and L.R.K.

Conflicts of interest

The authors declare that there are no conflicts of interest.

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